

ADYANSH GUPTA

Robotics Engineer — Perception, Navigation & Sensor Fusion

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ABOUT ME

Robotics Engineer currently pursuing an M.Sc. in Robotic Systems Engineering at RWTH Aachen. Specialized in bridging Perception/Computer Vision with Navigation/SLAM for Autonomous Mobile Robots (AMRs). Proven track record in optimizing complex C++ perception pipelines using HPC techniques and designing multi-sensor fusion systems. Passionate about transforming theoretical AI models into robust, real-world hardware deployments in dynamic environments.

EDUCATION

RWTH Aachen University

Robotic Systems Engineering (M.Sc.)

Oct 2025 – Present

Aachen, Germany

Vishwakarma Institute of Information Technology

Mechanical Engineering (BTech) - CGPA: 9.4/10 (German Grade Equivalent: 1.3)

Jan 2022 – June 2025

Pune, India

EXPERIENCE

Addverb Technologies

Mobile Robotics Intern

April 2025 – August 2025

Noida, India

- Engineered a new C++ implementation of the WhyCode/Whycon fiducial marker system, utilizing HPC techniques (AVX-512 SIMD) on an Intel Industrial PC to slash CPU utilization from 180-190% down to 10-15%.
- Worked on camera drivers using shared memory to provide the perception system with low-latency image data.
- Assisted in integrating and testing Lidar-Inertial Odometry (LIO) and IMU correction software to improve robot positioning accuracy.
- Conducted hands-on testing and validation of all developed software on physical robots to ensure reliable real-world performance.

Sakar Robotics

Robotics Intern

August 2024 – April 2025

Pune, India

- Collaborated on the development of autonomous industrial AMRs, focusing on advanced robotics and navigation systems for industrial applications.
- Implemented computer vision tasks using advanced AI technologies, including object detection and scene understanding.
- Worked extensively with NVIDIA Jetson hardware, leveraging Isaac ROS and GPU-accelerated packages to optimize robotic performance.
- Contributed to the software development of an all-terrain rover, gaining hands-on experience with advanced technologies to enhance reliability.

Avignon University

Drone Systems Development Internship

June 2024 – July 2024

Avignon, France

- Worked on the development of drone systems for Indoor Navigation, Object Tracking, and ArUco-based Inventory Management using ROS2.
- Achieved a perfect score of 20/20 for my performance, distinguishing myself among all students in the internship.

Dassault Systems

Engineering Design Internship Program

Sept 2023 – Jan 2024

Pune, India

- Led the conversion of a Pelton Wheel Turbine into a precise 3D digital twin.
- Performed the simulation of this turbine for optimal performance.

VIIT Robotics (College Club)

Vice President, Mechatronics Engineer

Aug 2022 – July 2024

Pune, India

- Led a team of talented robotics enthusiasts, fostering collaboration and ensuring efficient project execution.
- Utilized mechatronics engineering skills to design, prototype, and test robotics components.

PROJECTS

AI+Robotics Hackathon 2025 (RWTH Aachen) | Best Technical Implementation, Isaac Sim, MoveIt

- Integrated a robust perception pipeline enabling a robotic arm to pick specific objects while dynamically avoiding table obstacles.
- Generated a real-time 3D environment mesh using **NvBlox** and leveraged **NVIDIA FoundationPose** to identify target object poses and create 3D bounding boxes.
- Processed the environment mesh by subtracting target objects and interfaced with the **MoveIt** planning scene for collision-free path generation, validating the pipeline in **Isaac Sim** before physical deployment.

OLIVE: Optimization of Lidar, Inertial, Vision & Encoders [↗](#) | WIP, ROS2, Sensor Fusion

- Developing a graph-based, loosely coupled multi-sensor fusion backend to robustly localize a robot by integrating IMU, wheel odometry, LiDAR, visual odometry, and fiducial markers.
- Currently implementing the LiDAR odometry subsystem, laying the groundwork for a highly accurate state estimation pipeline in the ROS2 ecosystem.

R2-ABU Robocon 2024 [↗](#) | ROS2 Humble, Micro-ROS, YOLO, SLAM, Nav2

- Developed a four-wheel **mecanum drive omnidirectional robot**, integrating all hardware actuators and sensors using **Micro-ROS**.
- Implemented **ball and silo detection** with **YOLO**, **line following** using **LSA08 sensors**, and **autonomous alignment** utilizing **TF Luna**.

Kiwi Drive Robot [↗](#) | ROS2, Micro-ROS, EKF, Gazebo

- Developed a **three-wheeled kiwi drive robot** with omni wheels, utilizing **Micro-ROS** and **ESP32** for locomotion and odometry data. Simulated the robot in Gazebo with sensors and navigation algorithms.
- Implemented **sensor fusion** using an **IMU sensor** and encoders with **extended Kalman filters**, integrated a **CSI camera** for object detection (CNN trained on LVIS), and utilized **LiDAR** for autonomous navigation.

Vanguard [↗](#) | CNN, Image Contouring, A* Algorithm, QGIS

- Developed a **CNN classification model** with noise reduction and data augmentation techniques to classify images into five categories.
- Implemented **image contouring** using an overhead camera to detect images on a map, enabling a **line-following robot** to prioritize and navigate using the **A* Algorithm** for optimal pathfinding.

Development of Drone Systems | ROS2, Nav2, SLAM Toolbox, CV

- Developed autonomous navigation systems using the Crazyflie drone, equipped with MultiRanger and Flow decks, enabling precise indoor navigation via ROS2, Nav2, and SLAM Toolbox.
- Worked on the Tello EDU drone for various computer vision tasks, including object detection, object following, face recognition, and warehouse management applications.

AgroBot Development | ROS2, Hardware Acceleration, CV

- Designed and developed a four-wheel omni wheel (X-Drive) robot using Studica robotics kits, enabling versatile movement for agricultural tasks such as pick-and-place operations and spraying.
- Implemented hardware-accelerated drivers for sensors and actuators and integrated computer vision for task recognition and completion.

SKILLS

Robotics: Robot Operating System (ROS/ROS2), Micro-ROS, Perception, Computer Vision, Isaac ROS, Nav2, SLAM, MoveIt, Gazebo

Programming: C++, Python, HPC, CUDA, Git, Linux, Docker

Artificial Intelligence: Machine learning, Deep learning, Reinforcement Learning, Predictive modeling

CAD, CAM and CAE: Fusion 360, Catia, Solidworks, 3D Experience, Ansys (FEA and CFD)

PUBLICATIONS AND PATENTS

- **Publication:** Deep Learning for the Classification of War-Related Events
- **Publication:** Detection and Classification of Bleeding and Non-Bleeding Frames in Wireless Capsule Endoscopy
- **Patent:** Advanced 3D Printed Remote-Controlled All-Terrain Rocker-Bogie Robot
- **Patent:** An Omni Directional Mobile Robot

CERTIFICATIONS

- **IIT Madras, Intellipaat:** Advanced Certification in Data Science and AI
- **Udemy:** ROS2 MoveIt, Gazebo, Navigation and Advanced Concepts
- **Coursera:** Autodesk Fusion 360 CAD/CAM/CAE and Generative Design Specialization